

Python-Based LDA MCQs & Answers

1. Which Python library provides a built-in implementation of Linear Discriminant Analysis?

- ☐ A) NumPy
- ☐ B) Pandas
- ☐ C) Scikit-learn
- ☐ D) Matplotlib

Answer: C) Scikit-learn

2. Which class is commonly used to implement LDA in Scikit-learn?

- ☐ A) PCA
- ☐ B) LinearDiscriminantAnalysis
- ☐ C) LogisticRegression
- ☐ D) DecisionTreeClassifier

Answer: B) LinearDiscriminantAnalysis

3. What is the purpose of the fit() method in Scikit-learn's LDA?

- ☐ A) Visualize data
- ☐ B) Learn discriminant directions from labeled data
- ☐ C) Normalize features
- ☐ D) Predict labels

Answer: B) Learn discriminant directions from labeled data

4. Which method is used to reduce dimensionality after fitting an LDA model?

- ☐ A) transform()
- ☐ B) predict()
- ☐ C) score()
- ☐ D) classify()

Answer: A) transform()

5. In Python, which library is most suitable for computing eigenvalues and eigenvectors manually?

- ☐ A) OpenCV
- ☐ B) NumPy
- ☐ C) Flask
- ☐ D) Seaborn

Answer: B) NumPy

6. What will happen if the target labels y are not provided while fitting an LDA model?

- ☐ A) LDA will automatically cluster data
- ☐ B) LDA cannot be trained properly
- ☐ C) LDA converts to PCA
- ☐ D) No effect

Answer: B) LDA cannot be trained properly

7. Which command is used to import LDA from Scikit-learn?

- ☐ A) from sklearn.decomposition import LDA
- ☐ B) from sklearn.lda import LDA
- ☐ C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
- ☐ D) from sklearn.linear_model import LDA

Answer: C) from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

8. What is the output shape after applying LDA with $n_components=2$ to a dataset containing 100 samples?

- ☐ A) (2,100)
- ☐ B) (100,2)
- ☐ C) (100,100)
- ☐ D) (2,2)

Answer: B) (100,2)

9. Which NumPy function is commonly used to compute eigenvalues and eigenvectors?

- ☐ A) np.mean()
- ☐ B) np.linalg.eig()
- ☐ C) np.dot()
- ☐ D) np.var()

Answer: B) np.linalg.eig()

10. In an LDA workflow, why is feature scaling often performed before applying LDA?

- ☐ A) To reduce memory usage
- ☐ B) To ensure features contribute comparably
- ☐ C) To increase sample size
- ☐ D) To remove labels

Answer: B) To ensure features contribute comparably

11. Which parameter controls the number of discriminant components in Scikit-learn LDA?

- ☐ A) `n_clusters`
- ☐ B) `n_neighbors`
- ☐ C) `n_components`

o D) max_depth

Answer: C) n_components

12. Which matrix operation is required when implementing LDA from scratch?

o A) Matrix inversion

o B) Image convolution

o C) FFT

o D) Hashing

Answer: A) Matrix inversion

13. While implementing LDA manually, which scatter matrix represents class compactness?

o A) Between-Class Scatter Matrix

o B) Within-Class Scatter Matrix

o C) Covariance Matrix

o D) Correlation Matrix

Answer: B) Within-Class Scatter Matrix

14. Which step comes first when implementing LDA from scratch?

o A) Compute eigenvectors

o B) Compute class means

o C) Train classifier

o D) Plot decision boundary

Answer: B) Compute class means

15. After computing eigenvectors in manual LDA, what should be done next?

o A) Randomly shuffle them

o B) Sort them according to eigenvalues

o C) Delete negative values

o D) Normalize labels

Answer: B) Sort them according to eigenvalues

16. Which NumPy function can be used to project data onto selected discriminant vectors?

o A) np.dot()

o B) np.max()

o C) np.sum()

o D) np.argmax()

Answer: A) np.dot()

17. What is the primary purpose of `lda.predict(X_test)`?

o A) Reduce dimensions

o B) Classify unseen samples

- o C) Scale data
- o D) Compute covariance

Answer: B) Classify unseen samples

18. Which metric is commonly used to evaluate an LDA classifier?

- o A) Accuracy
- o B) RMSE
- o C) MAPE
- o D) MAE

Answer: A) Accuracy

19. In Scikit-learn, LDA can be used for:

- o A) Classification only
- o B) Dimensionality reduction only
- o C) Both classification and dimensionality reduction
- o D) Clustering only

Answer: C) Both classification and dimensionality reduction

21. You are asked to implement Linear Discriminant Analysis from scratch using only NumPy (without using sklearn.discriminant_analysis.LinearDiscriminantAnalysis). Which sequence of steps is correct?

- A) Standardize Data → Compute Class Means → Compute Within-Class Scatter Matrix → Compute Between-Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$ → Select Top Eigenvectors → Project Data
- B) Compute FFT → Compute Covariance Matrix → Apply K-Means → Project Data
- C) Compute PCA Components → Apply Logistic Regression → Predict Labels
- D) Normalize Labels → Compute Decision Tree → Extract Features → Project Data

Answer: A) Standardize Data → Compute Class Means → Compute Within-Class Scatter Matrix → Compute Between-Class Scatter Matrix → Solve Eigenvalue Problem of $S_W^{-1}S_B$ → Select Top Eigenvectors → Project Data

22. What is the purpose of the following line in a manual LDA implementation?
eigvals, eigvecs = np.linalg.eig(np.linalg.inv(Sw).dot(Sb))

- A) Compute principal components
- B) Compute discriminant directions
- C) Compute cluster centroids
- B) Compute feature scaling factors

Answer: B) Compute discriminant directions

23. Consider:

$X_{lda} = X \cdot W$

What does W represent?

- A) Weight initialization matrix
- B) Matrix of selected discriminant vectors
- C) Covariance matrix
- D) Distance matrix

Answer: B) Matrix of selected discriminant vectors

24. In a scratch implementation, why are eigenvectors sorted in descending order of eigenvalues?

- A) To reduce execution time
- B) To select directions with maximum discriminative power
- C) To normalize data
- D) To balance classes

Answer: B) To select directions with maximum discriminative power

25. When implementing LDA manually, what is the most likely issue if S_w is singular?

- A) Labels disappear
- B) Matrix inversion fails
- C) Data becomes nonlinear
- D) Eigenvalues become complex automatically

Answer: B) Matrix inversion fails